

CSE 311:

Data Communication

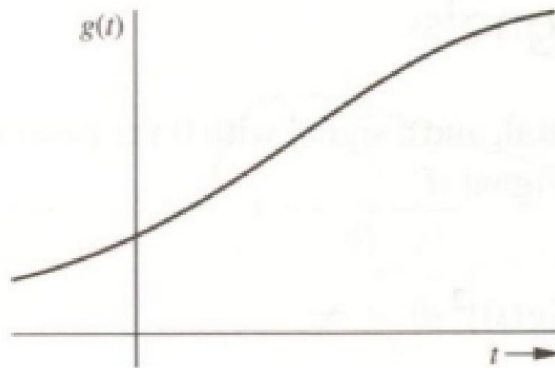
Instructor:
Dr. Md. Monirul Islam

Classification of Signals

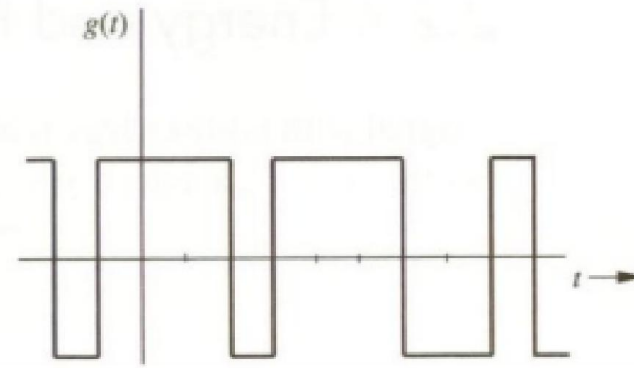
Continuous amplitude

Discrete amplitude

Continuous
time

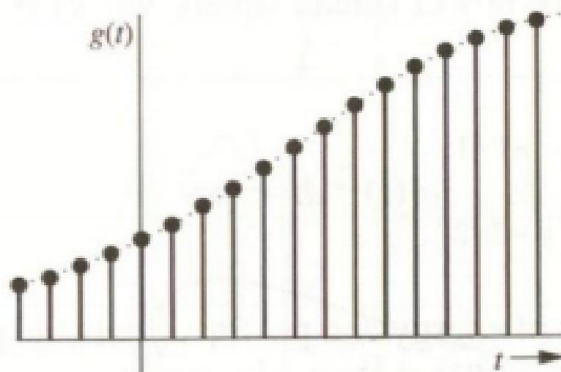


(a)

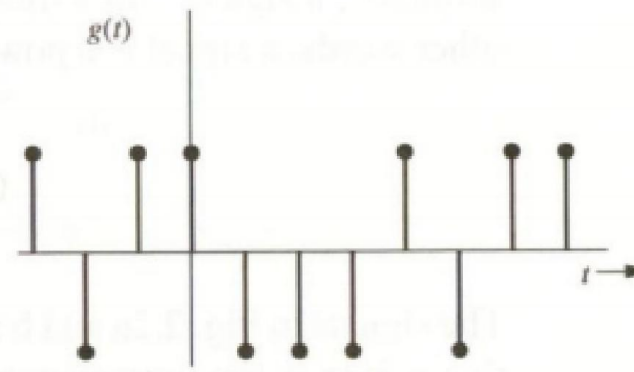


(b)

Discrete time



(c)



(d)

Analog and Digital Signal

Analog Signal

- Continuous amplitude, i.e., takes any value in a continuous range.
- May be both continuous and discrete time.

Digital Signal

- Discrete amplitude, i.e., amplitude can take only a finite number of values.
- Values need not be always integer.
- Not necessarily always binary, rather M-ary.
- May be both continuous and discrete time.

Analog and Digital Signal: Examples

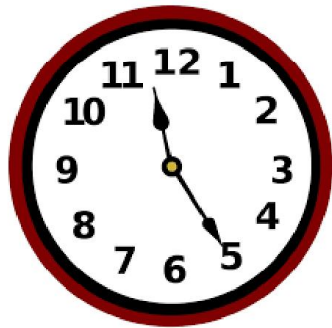
Analog

Digital

Thermometer



Clock



Blood Pressure
Monitor



Benefits of Digital Message/Signal

1. Quality, e.g., enhanced noise immunity
2. Economics

Benefits of Digital Message/Signal

1. noise immunity

- Represented by binary or M-ary pulses

Sent



Benefits of Digital Message/Signal

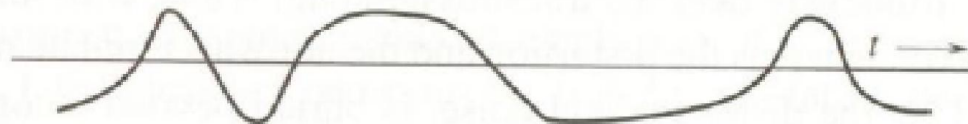
1. noise immunity

- Represented by binary or M-ary pulses

Sent



Received
w/o noise

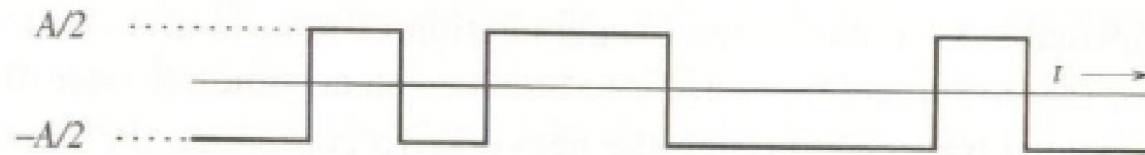


Benefits of Digital Message/Signal

1. noise immunity

- Represented by binary or M-ary pulses

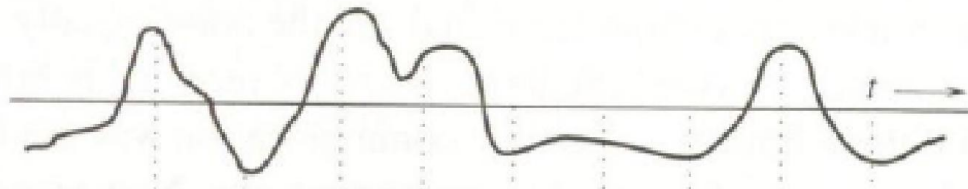
Sent



Received
w/o noise



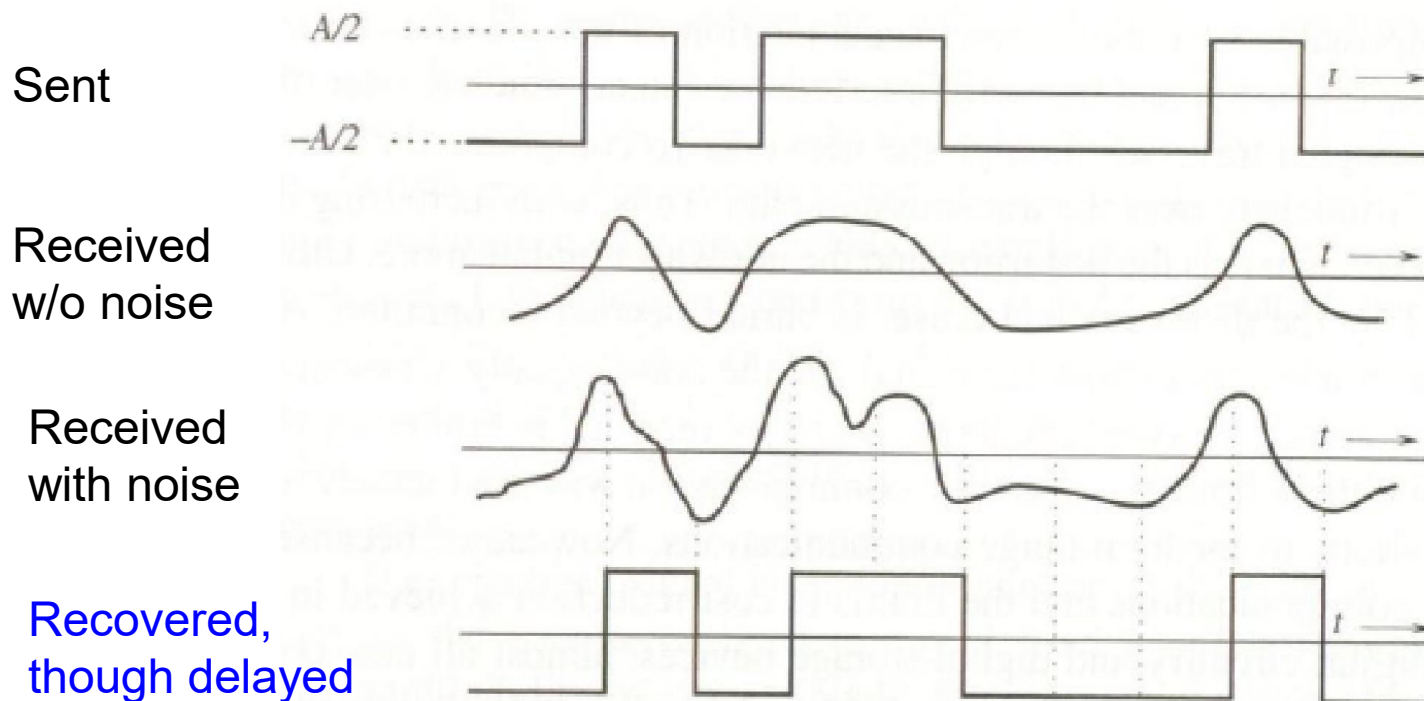
Received
with noise



Benefits of Digital Message/Signal

1. noise immunity

- Recovered despite small distortion and noises

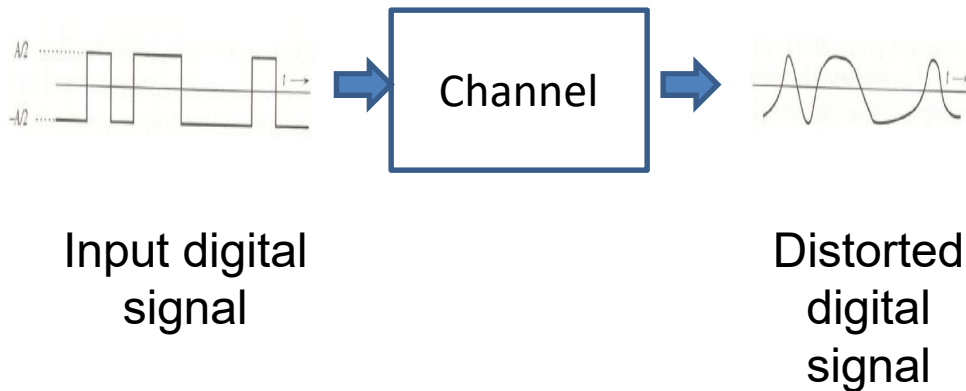


Repeater's Role in Digital Message/Signal

- Distortion and noise are unavoidable in channel
- Repeaters and nodes regenerates digital pulses

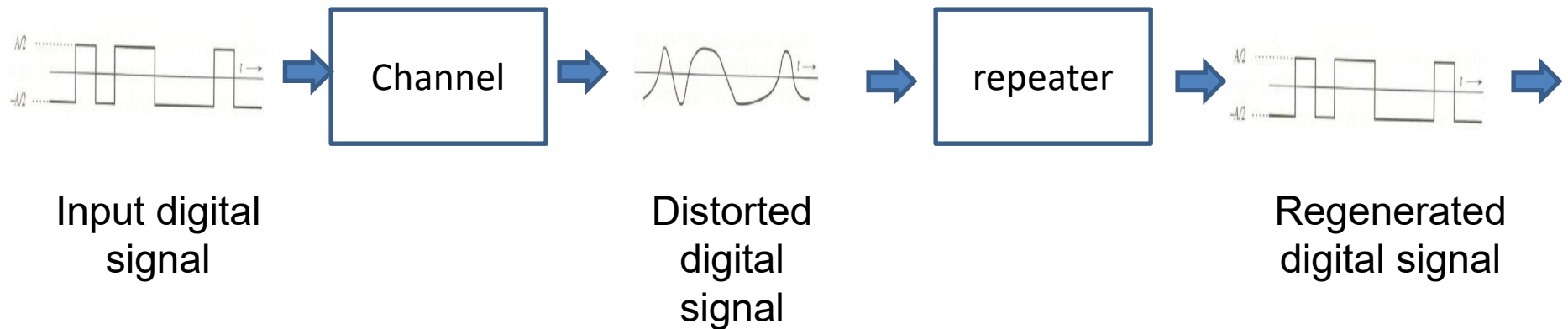
Repeater's Role in Digital Message/Signal

- Distortion and noise are unavoidable in channel
- Repeaters and nodes regenerates digital pulses



Repeater's Role in Digital Message/Signal

- Distortion and noise are unavoidable in channel
- Repeaters and nodes regenerates digital pulses

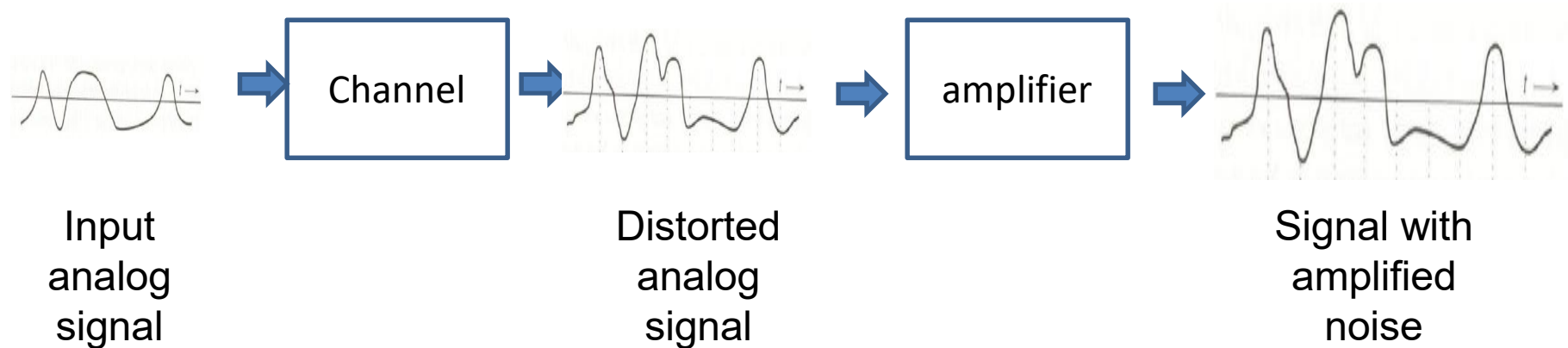


Repeater's Role in Digital Message/Signal

- Distortion and noise are unavoidable in channel
- Repeaters are filters and amplifiers in analog signals
- Amplifier amplifies both signal and noise

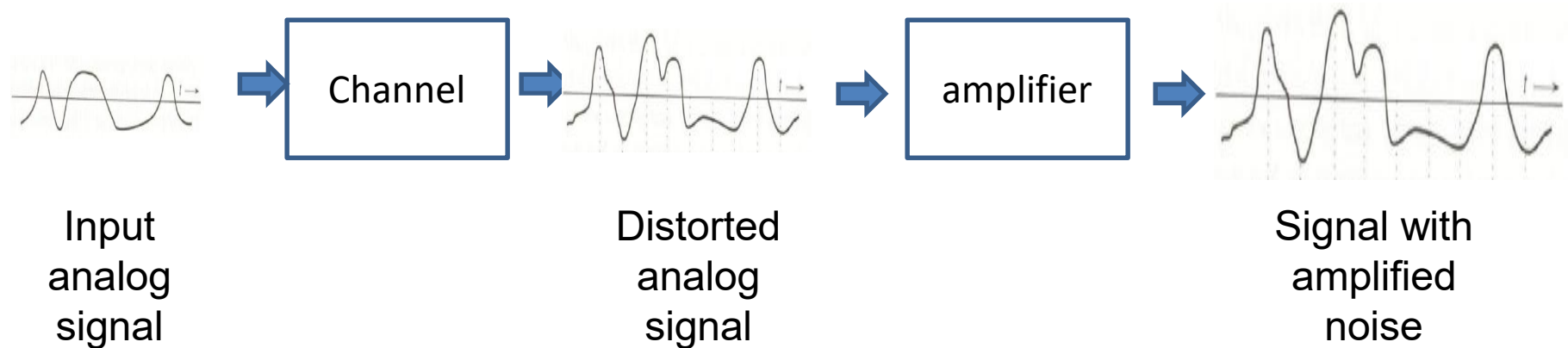
Repeater's Role in Digital Message/Signal

- Distortion and noise are unavoidable in channel
- Repeaters are filters and amplifiers in analog signals
- Amplifier amplifies both signal and noise



Repeater's Role in Digital Message/Signal

- Distortion and noise are unavoidable in channel
- Repeaters are filters and amplifiers in analog signals
- Amplifier amplifies both signal and noise



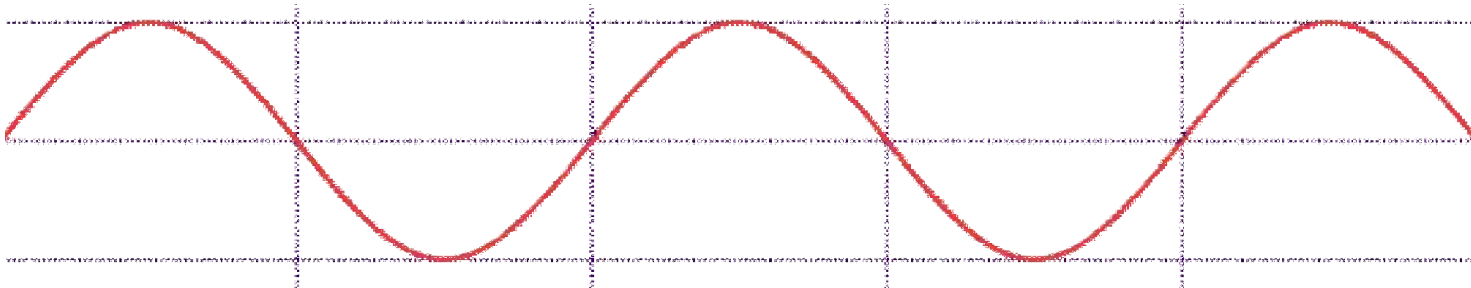
Noise accumulates along the path! No improvement at all !!

Analog to Digital Conversion of Message/Signal

- A/D conversion enables digital communication to convey analog signals

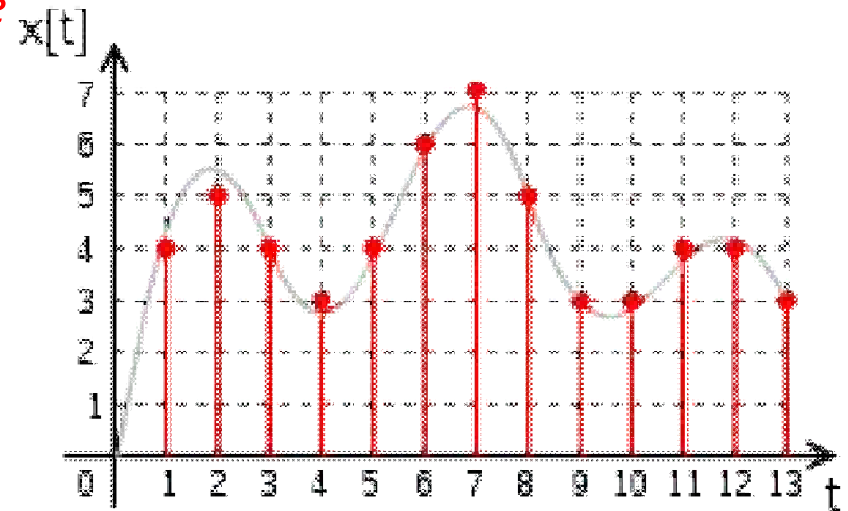
Analog to Digital Conversion of Message/Signal

- A/D conversion enables digital communication to convey analog signals
- Analog signal characteristics
 - values are continuous
 - defined over continuous/discrete time



Analog to Digital Conversion of Message/Signal

- A/D conversion enables digital communication to convey analog signals
- Digital signal characteristic's
 - values are a *finite discrete* set
 - defined over *preferably discrete time*

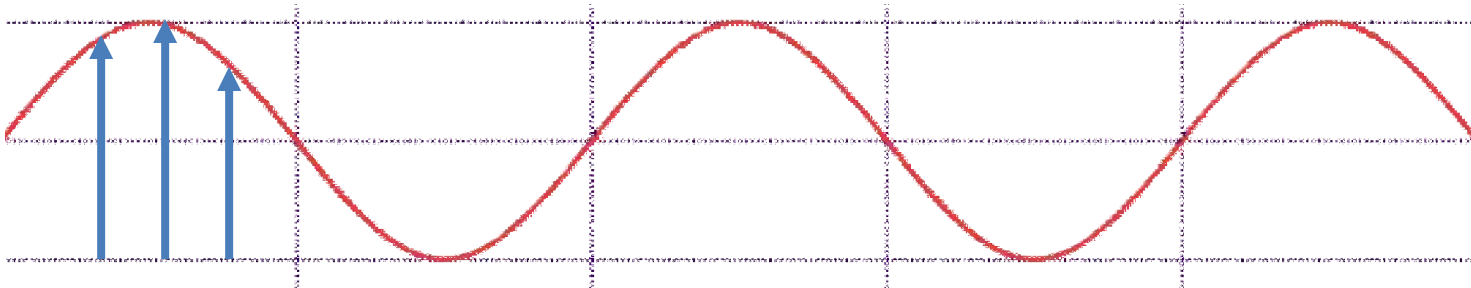


Analog to Digital Conversion of Message/Signal

- 2 major steps
 - Sampling
 - Quantization

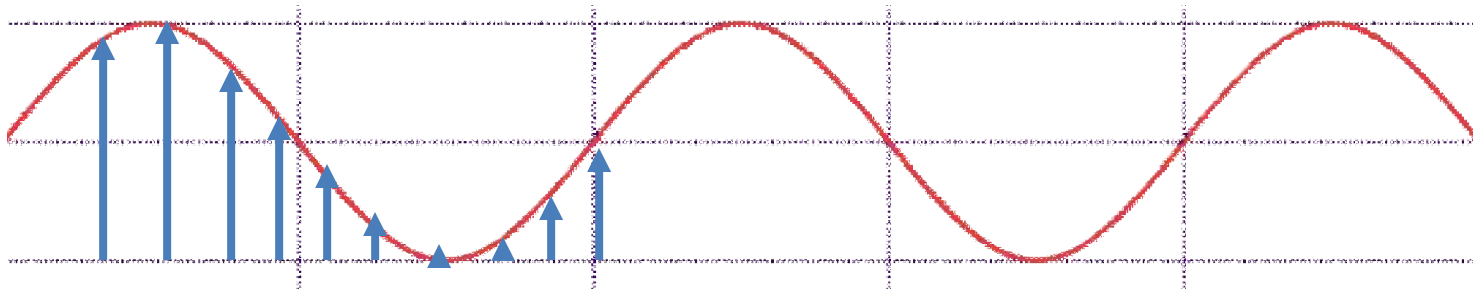
Analog to Digital Conversion of Message/Signal

- Sampling
 - Governed by Nyquist 's Sampling theory
 - Selects points for sampling



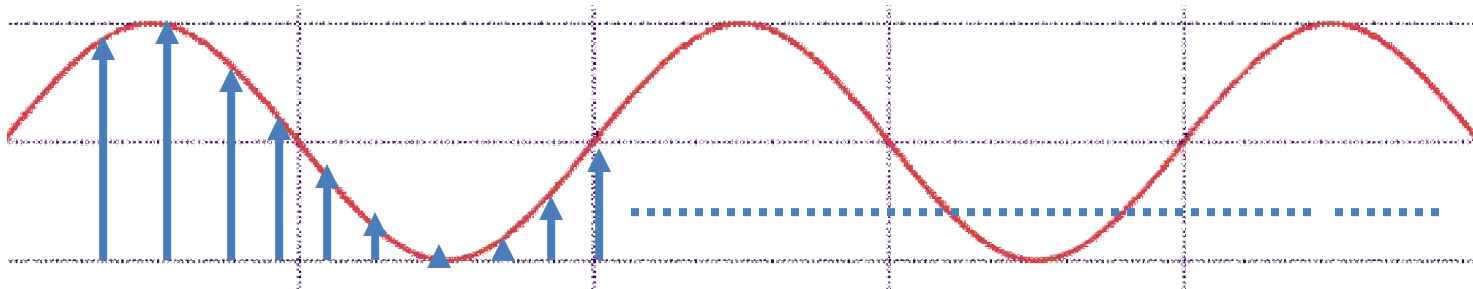
Analog to Digital Conversion of Message/Signal

- Sampling
 - Governed by Nyquist 's Sampling theory
 - Selects points for sampling



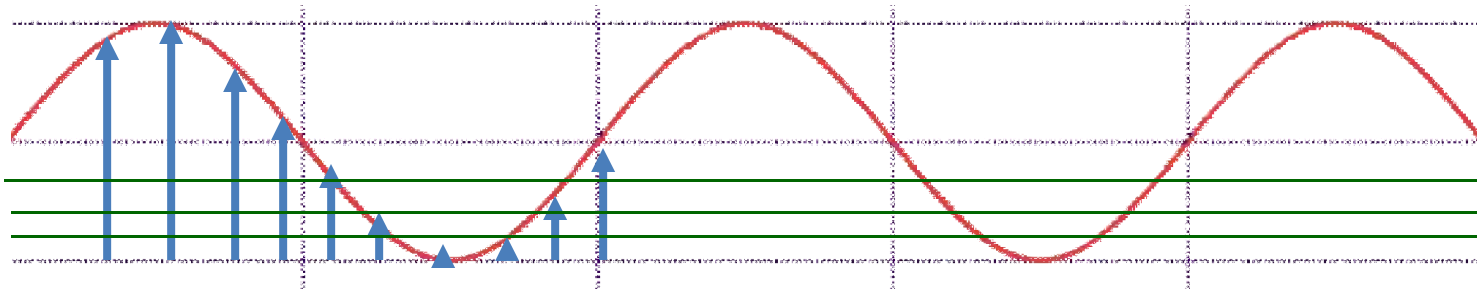
Analog to Digital Conversion of Message/Signal

- Sampling
 - Governed by Nyquist 's Sampling theory
 - Selects points for sampling



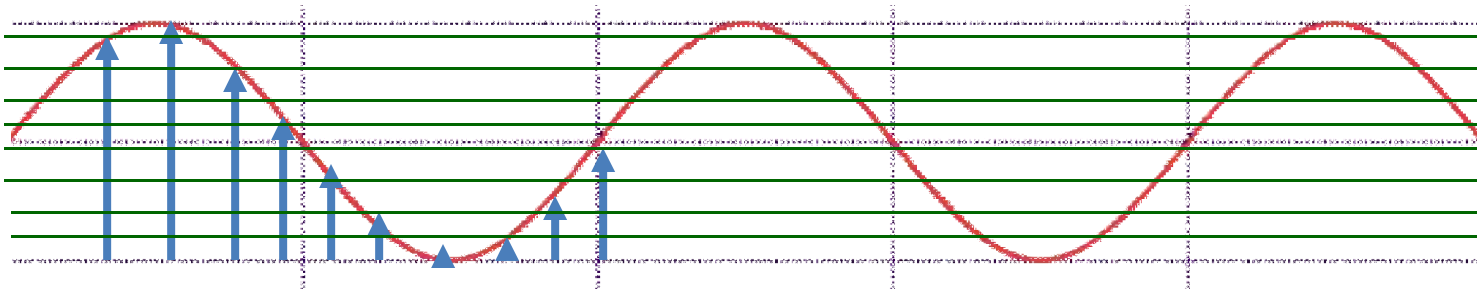
Analog to Digital Conversion of Message/Signal

- Quantization
 - Values replaced by a set of L distinct values
 - Usually $L = 2^k$



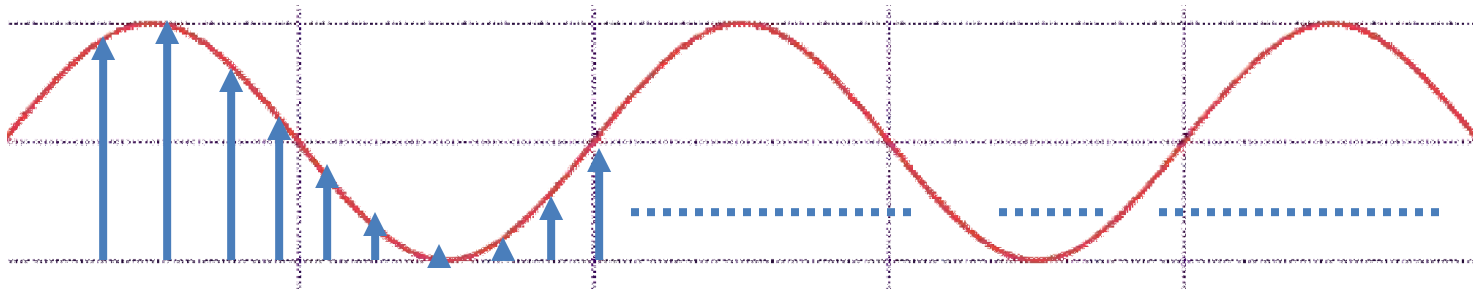
Analog to Digital Conversion of Message/Signal

- Quantization
 - Values replaced by a set of L distinct values
 - Usually $L = 2^k$



Analog to Digital Conversion of Message/Signal

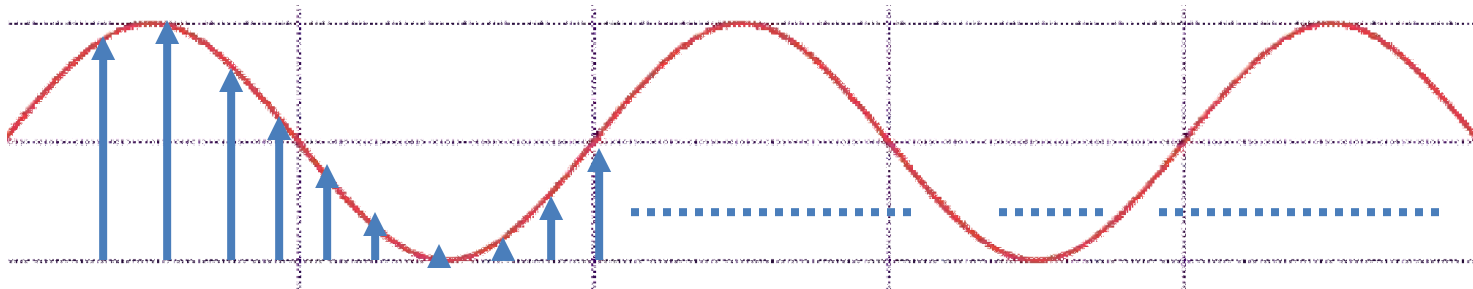
- Increasing sampling rate retains original shape



Analog to Digital Conversion of Message/Signal

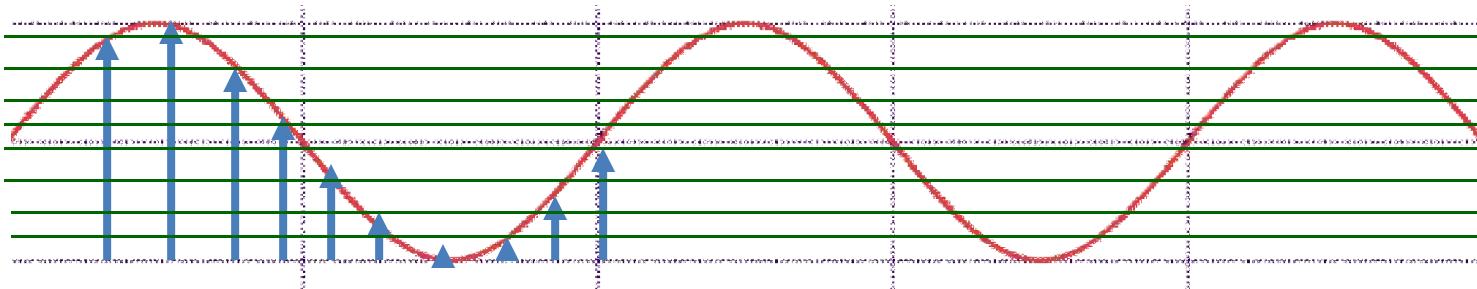
- Increasing sampling rate retains original shape

Remember
Nyquist's theorem!



Analog to Digital Conversion of Message/Signal

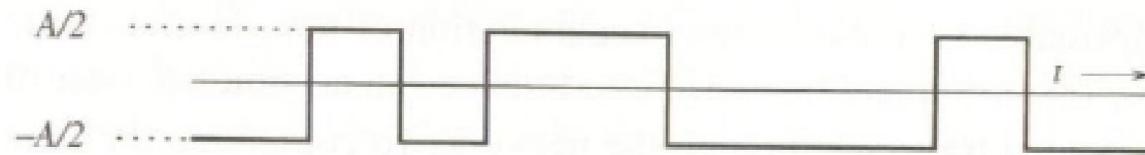
- Increasing Quantization level L
 - increases accuracy
 - more noise immunity
 - but requires higher channel bandwidth



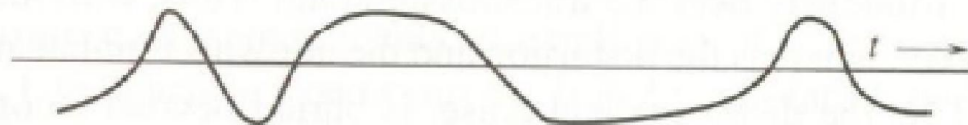
Analog to Digital Conversion of Message/Signal

Recall this figure

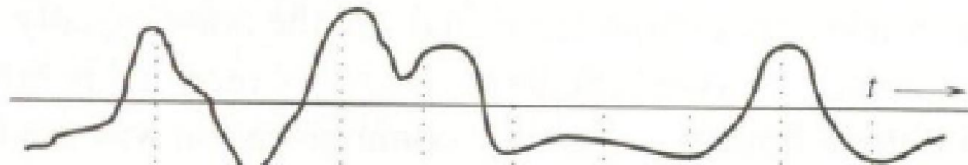
Sent



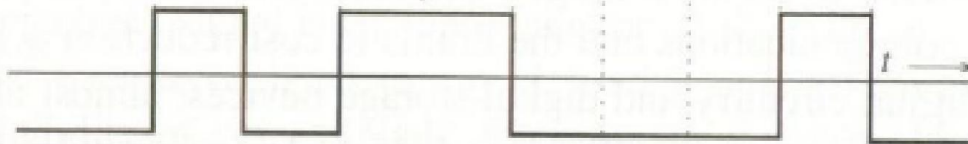
Received
w/o noise



Received
with noise



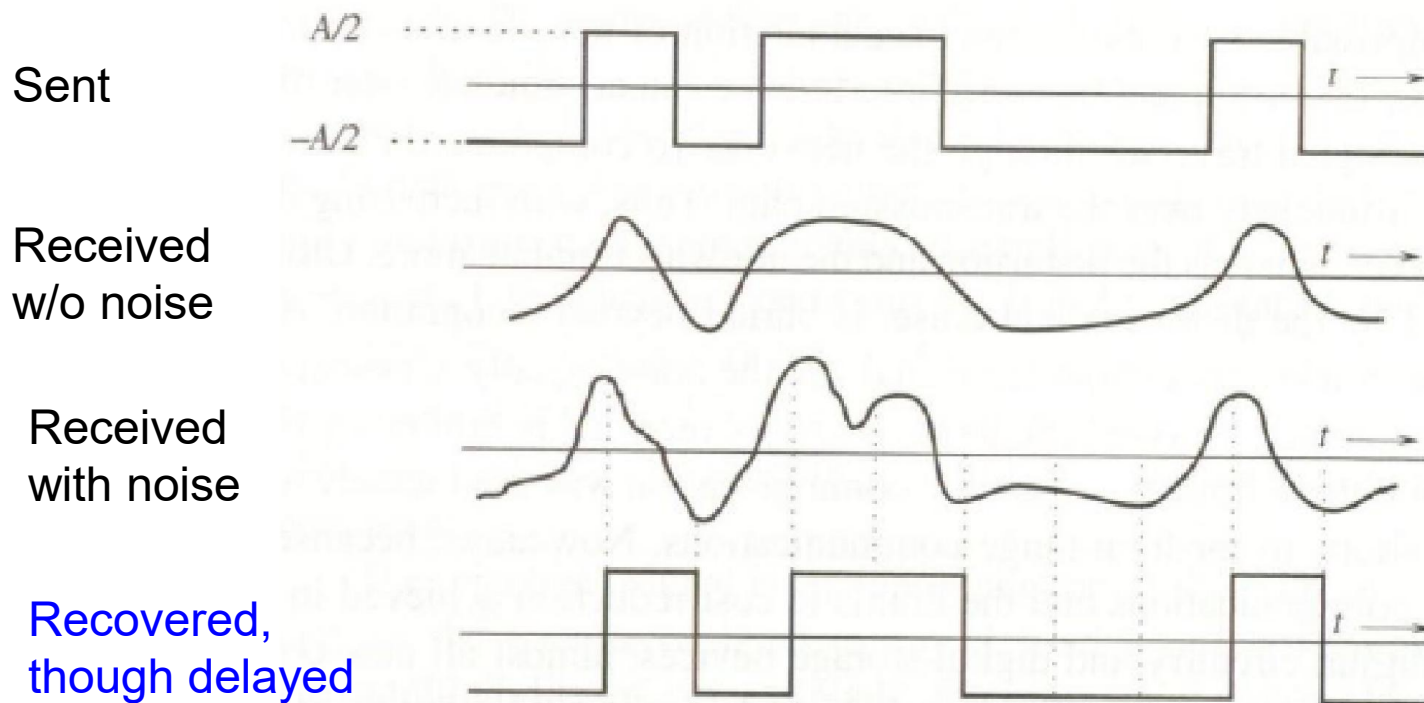
Recovered,
though delayed



Analog to Digital Conversion of Message/Signal

- Detection is easy when $A \gg \text{noise}$
- Usually $A \gg 5\text{-}10$ times of noise

Recall this figure



Analog to Digital Conversion of Message/Signal

- Problem: quantization error is unavoidable

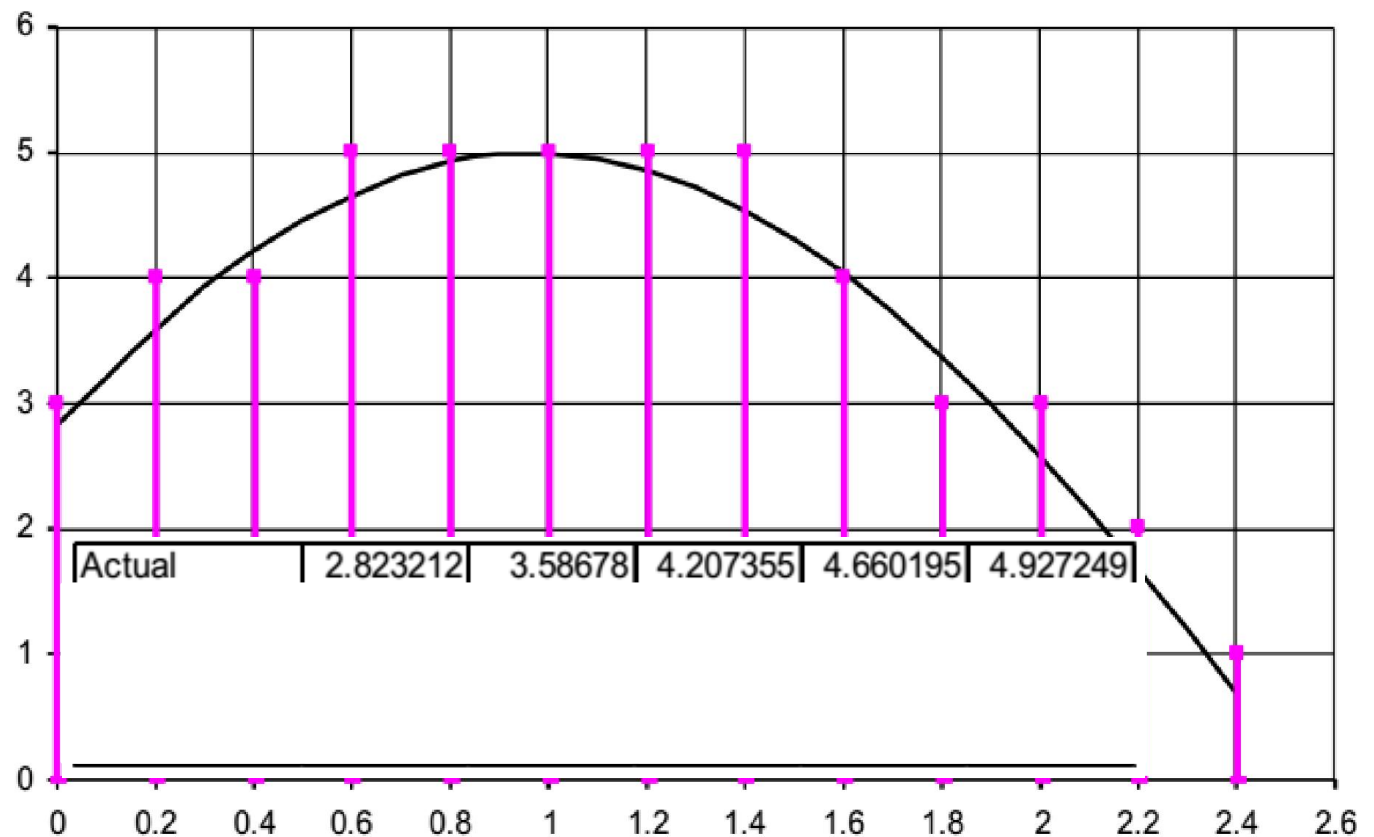
Analog to Digital Conversion of Message/Signal

- Problem: quantization error is unavoidable



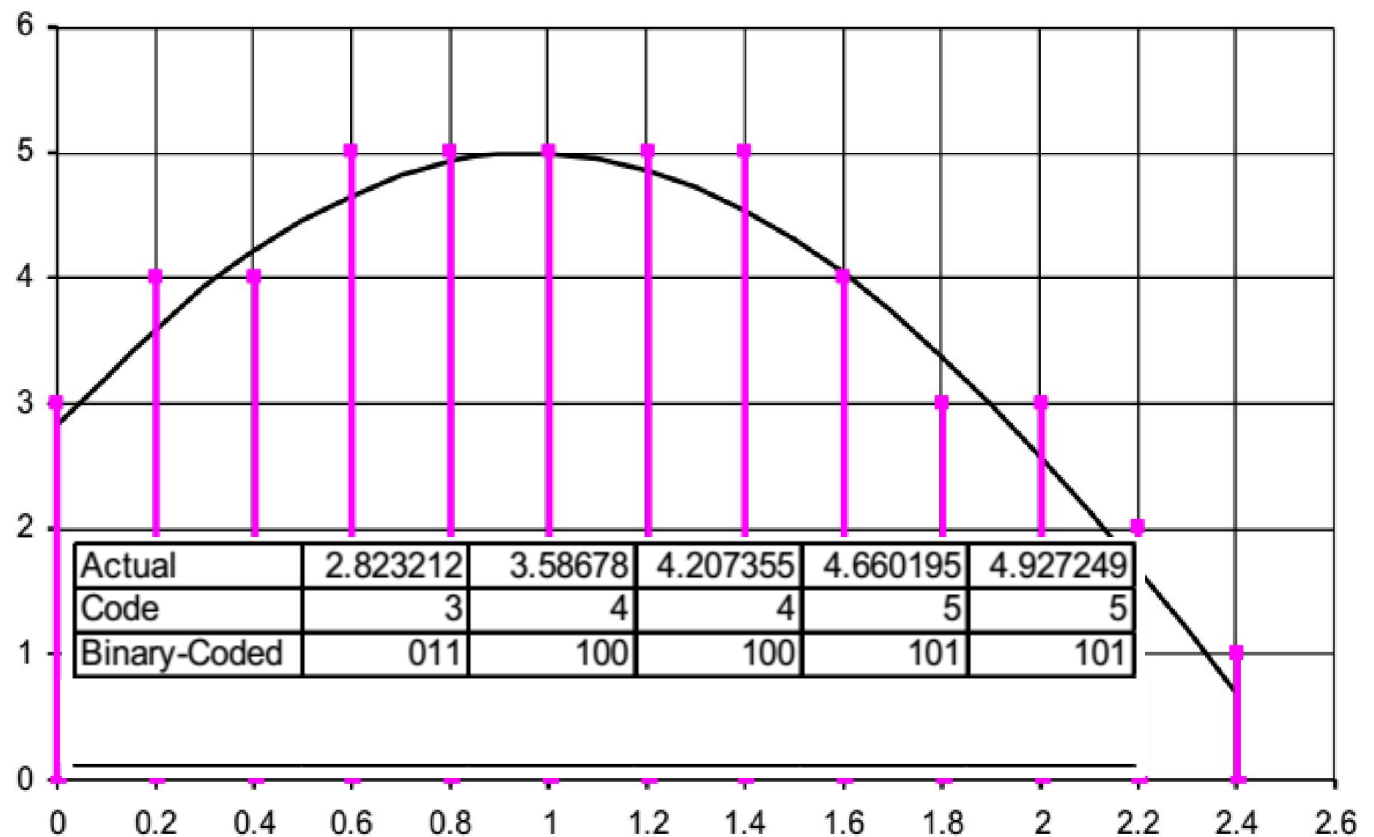
Analog to Digital Conversion of Message/Signal

- Problem: quantization error is unavoidable



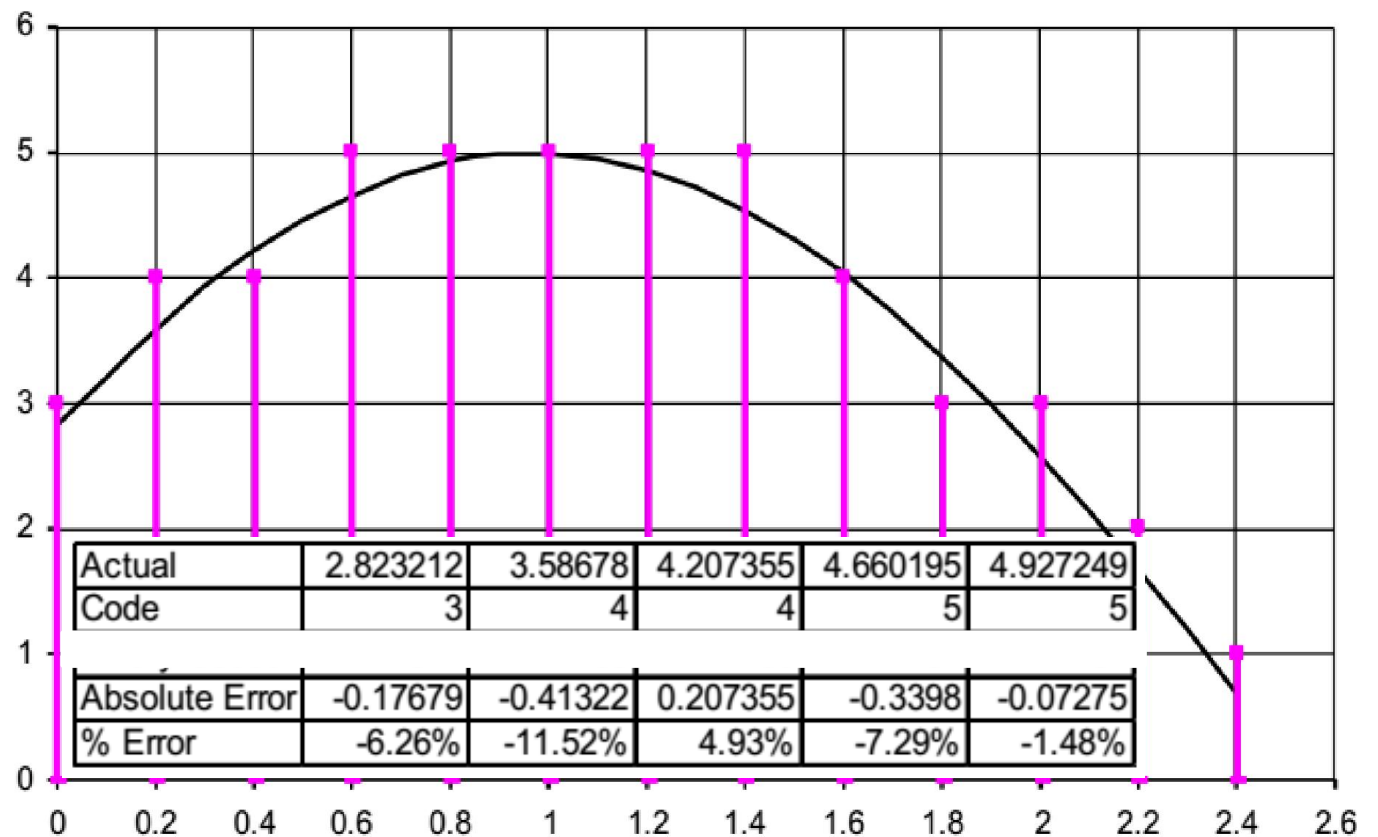
Analog to Digital Conversion of Message/Signal

- Problem: quantization error is unavoidable



Analog to Digital Conversion of Message/Signal

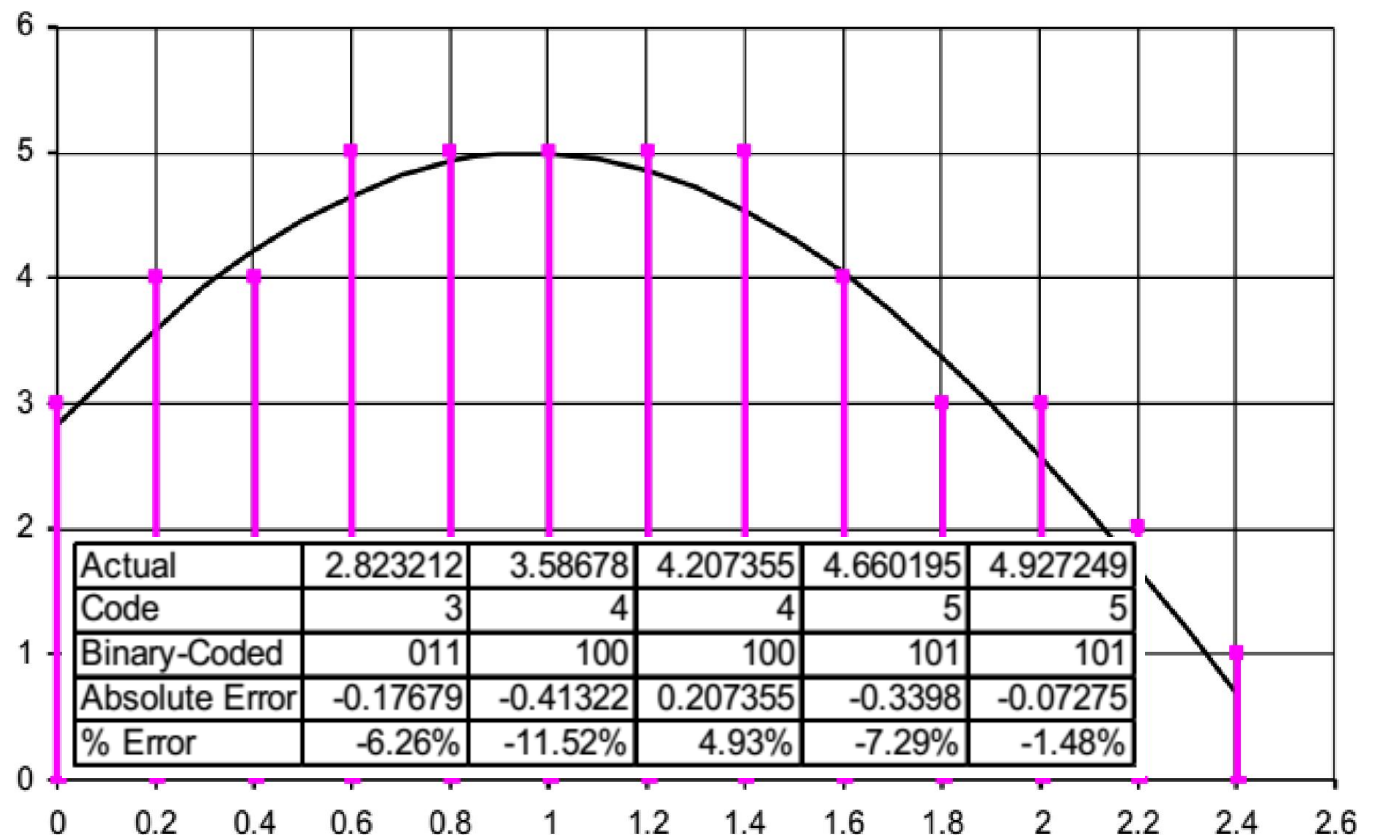
- Problem: quantization error is unavoidable



Analog to Digital Conversion of Message/Signal

- Problem: quantization error is unavoidable

Quantization error can be minimized increasing L

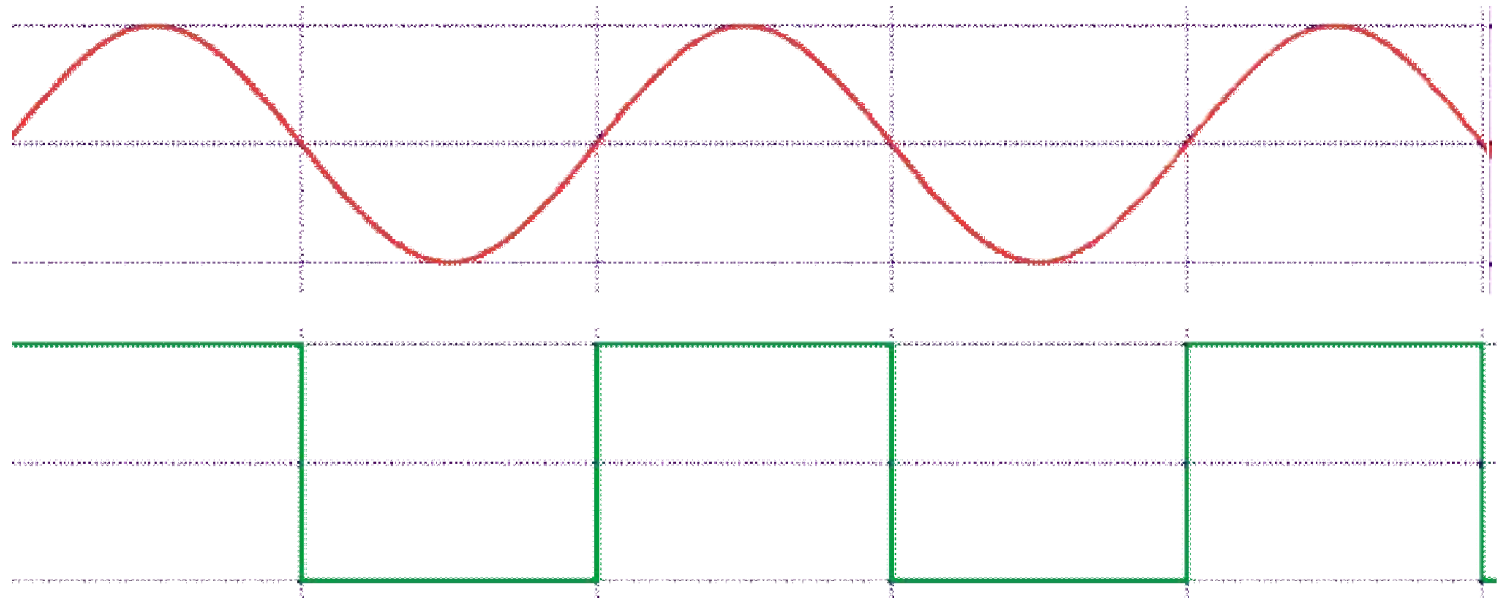


Representation of Digital Signal: Pulse Coded Modulation

- Assume, No. of quantization level, $L = 2$

Representation of Digital Signal: Pulse Coded Modulation

- Assume, No. of quantization level, $L = 2$
 - Easy to represent or transmit



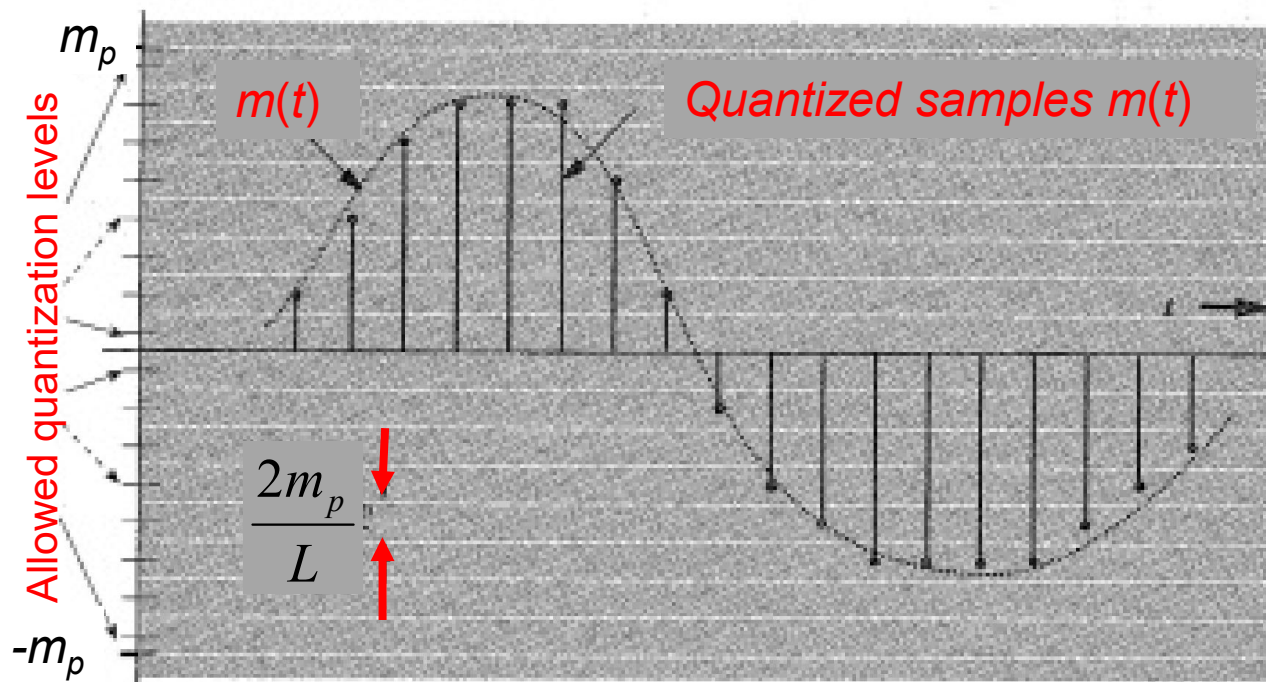
Representation of Digital Signal: Pulse Coded Modulation

- If No. of quantization level, $L \gg 2$

Representation of Digital Signal: Pulse Coded Modulation

- If No. of quantization level, $L \gg 2$

Each sample is represented by one of L levels

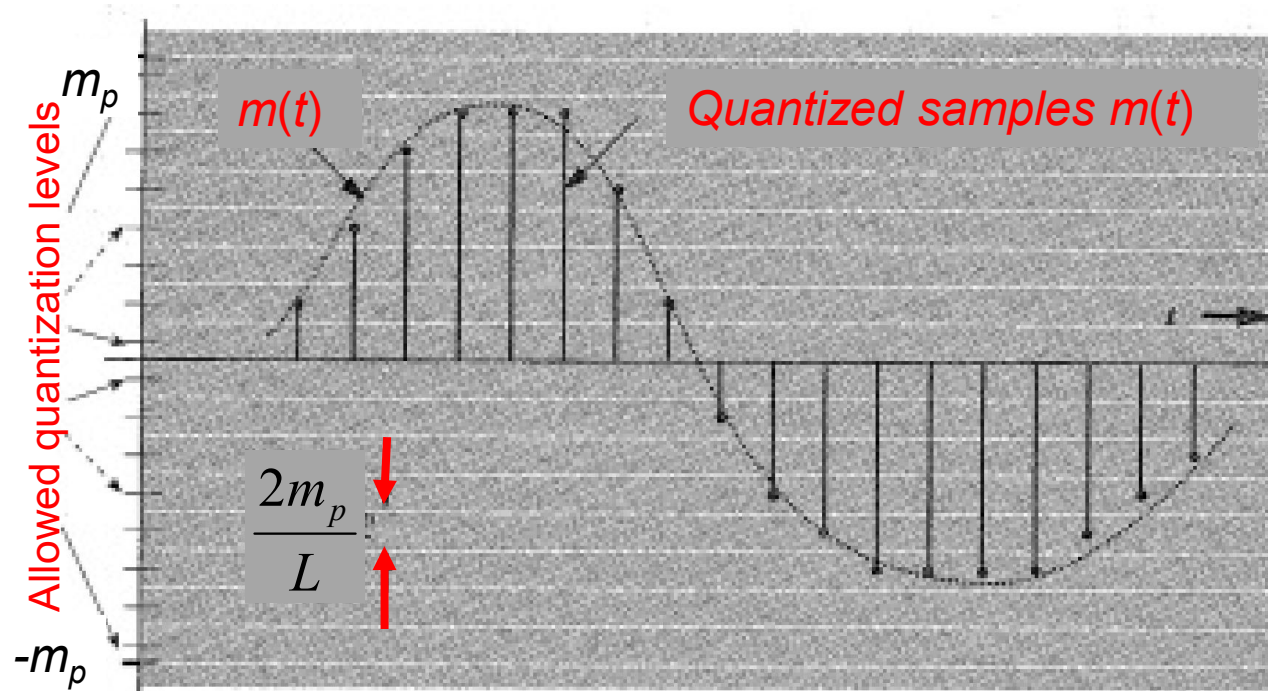


Representation of Digital Signal: Pulse Coded Modulation

- If No. of quantization level, $L \gg 2$
 - Solution is PCM
 - Each quantized value is represented by a *sequence of binary pulses*.

Representation of Digital Signal: Pulse Coded Modulation

- Assume, $L = 16$



Representation of Digital Signal: Pulse Coded Modulation

- Assume, $L = 16$
 - Each quantized value is represented by a *sequence of FOUR binary pulses*.

Digit	Binary equivalent	Pulse code waveform
0	0000	
1	0001	
2	0010	
3	0011	
4	0100	
5	0101	
6	0110	
7	0111	
8	1000	
9	1001	
10	1010	
11	1011	
12	1100	
13	1101	
14	1110	
15	1111	